COMPASS TEMPEST Sulfide

June-August 2026 Samples

2026-08-28

Code Set up

run information

```
##Before or After Reruns (YES or NO)
After_Reruns = "YES"
#If NO will not write final data file
  #And Will Write a Rerun File

###things that need to be changed
sample_month = "June-August"
sample_year = "2026"
Run_Date = as.Date("08/27/2026", format = "%m/%d/%Y")
Run_by = "Zoe Read, Melanie Giessner"#Instrument user
Script_run_by ="Zoe Read"#Code user

Run_notes="All dups and related samples had values of 0. 3/3 spikes were bad. "#any notes from run

#Plate Data
folder_path_rawdata <- paste0("Raw Data")
Raw_plates <- list.files(path = folder_path_rawdata, full.names = TRUE,pattern = "Plate\\d+")
(Raw_plates)
```

```
## [1] "Raw Data/TEMPEST_JuneAugust2026_Plate1_20260827.csv"
## [2] "Raw Data/TEMPEST_JuneAugust2026_Plate2_20260827.csv"
## [3] "Raw Data/TEMPEST_JuneAugust2026_Plate3_20260827.csv"
```

```
#Sample ID Path
# SampleID_path = "Raw Data/TEMPEST_SO4_20260501-20260612_corrected.csv"

# Define the file path for QAQC log file - NO Need to change just check year
log_path <- "Raw Data/Sulfide_STD_QAQC.csv"

# Define final path for data be sure to change year and month and run number
final_path <- "Processed Data/COMPASS_TEMPEST_H2S_202606-202608_processed.csv"
rerun_path <- "Processed Data/COMPASS_TEMPEST_H2S_202606-202608_reruns.csv"
```

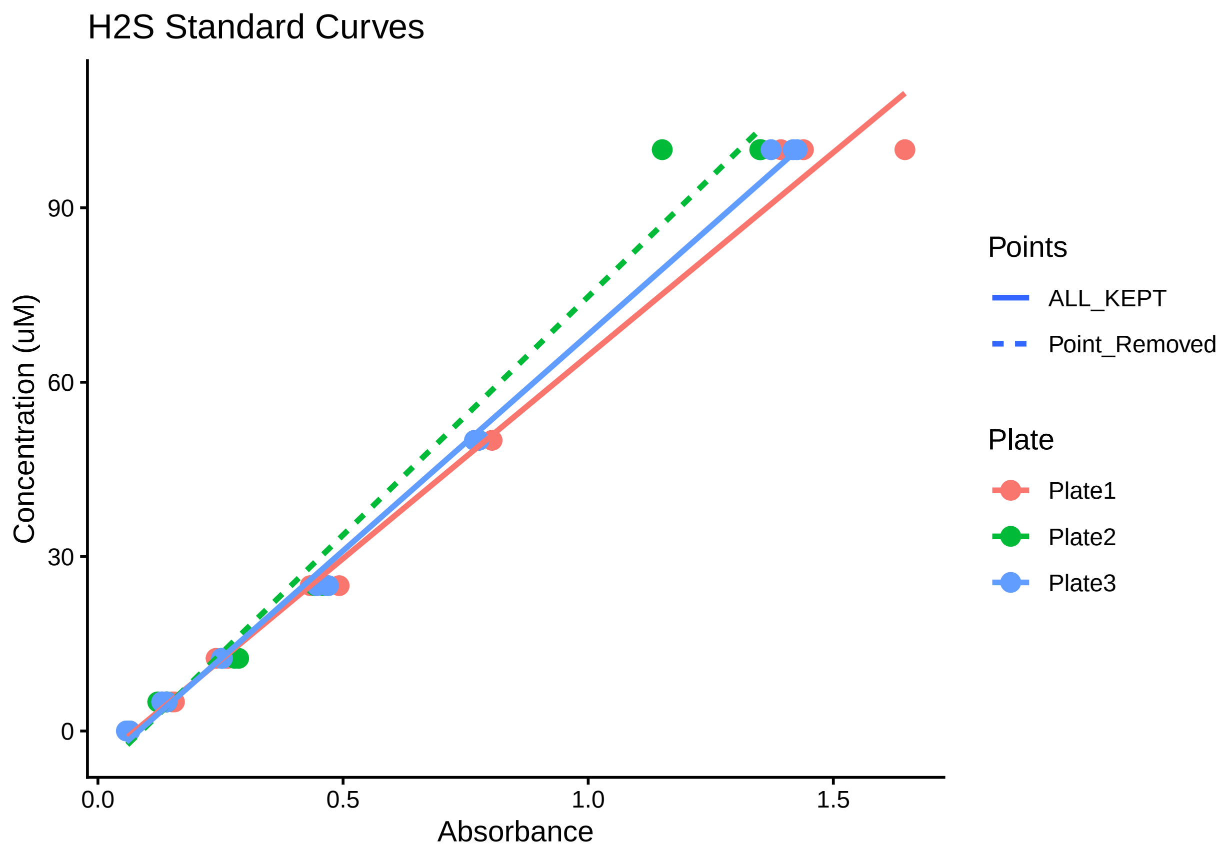
##Read in data

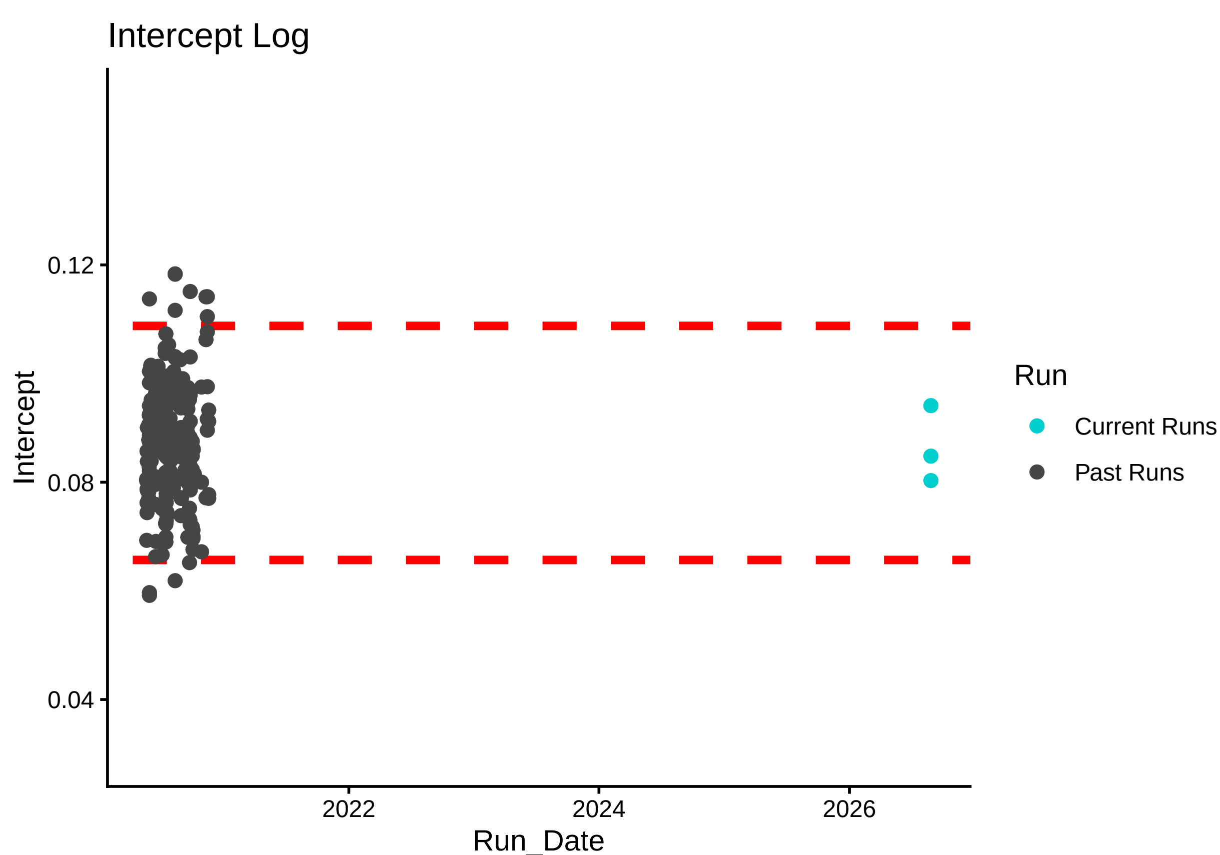
##Remove High CV Stds

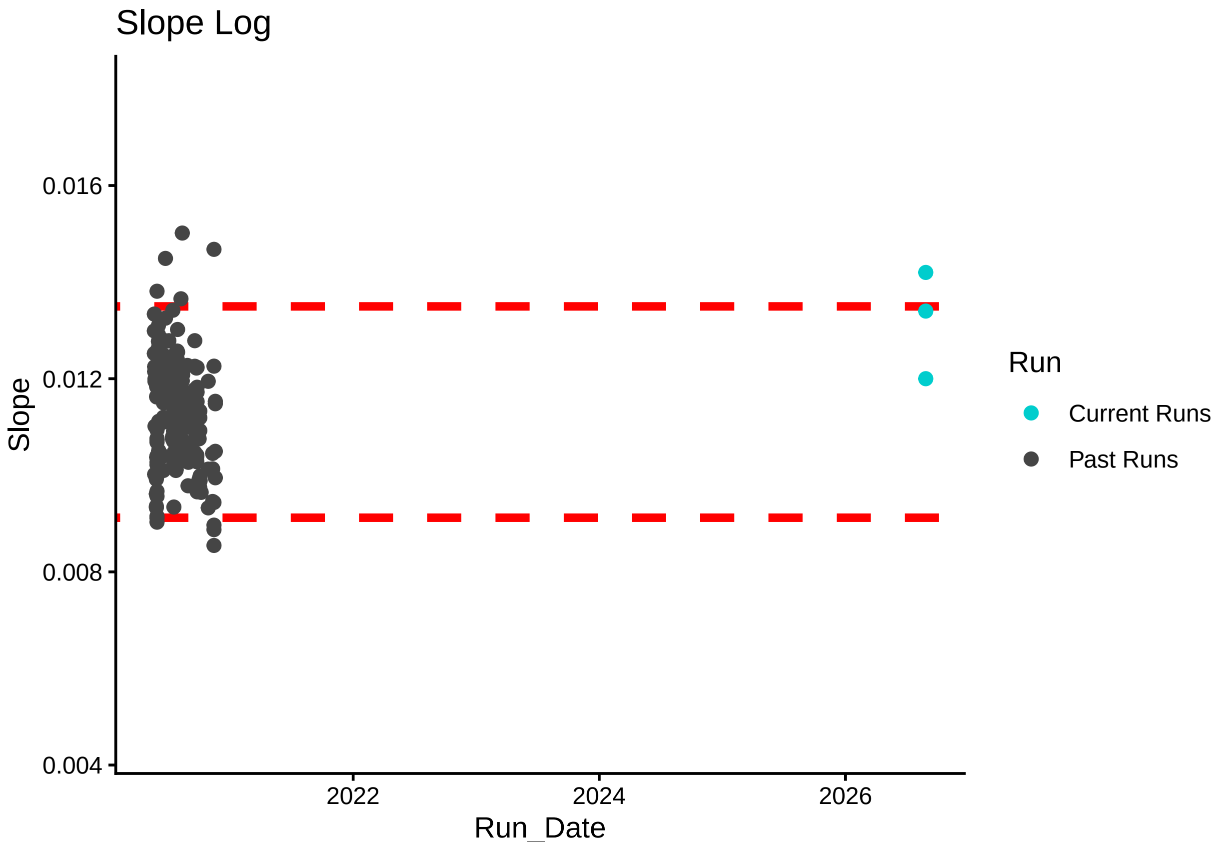
##Check R2 & Remove Standards if Necessary

```
## Joining with ‘by = join_by(Plate, IDs)‘
```

Plot Standard Curves





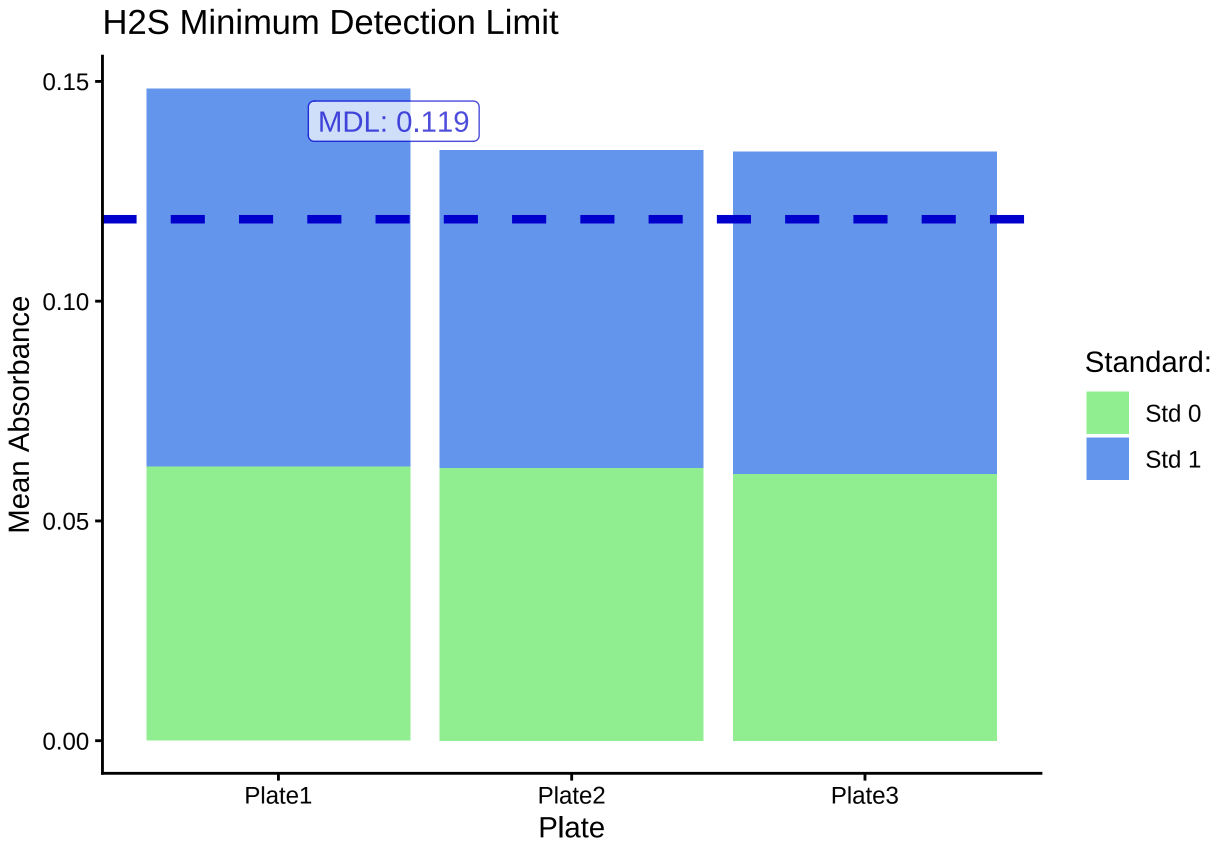


Return Flags For Standard Curve

Plate	R2_Flag	CV_Flag
Plate1	R2_Good	CV_Good
Plate2	R2_Bad	CV_Good
Plate3	R2_Good	CV_Good

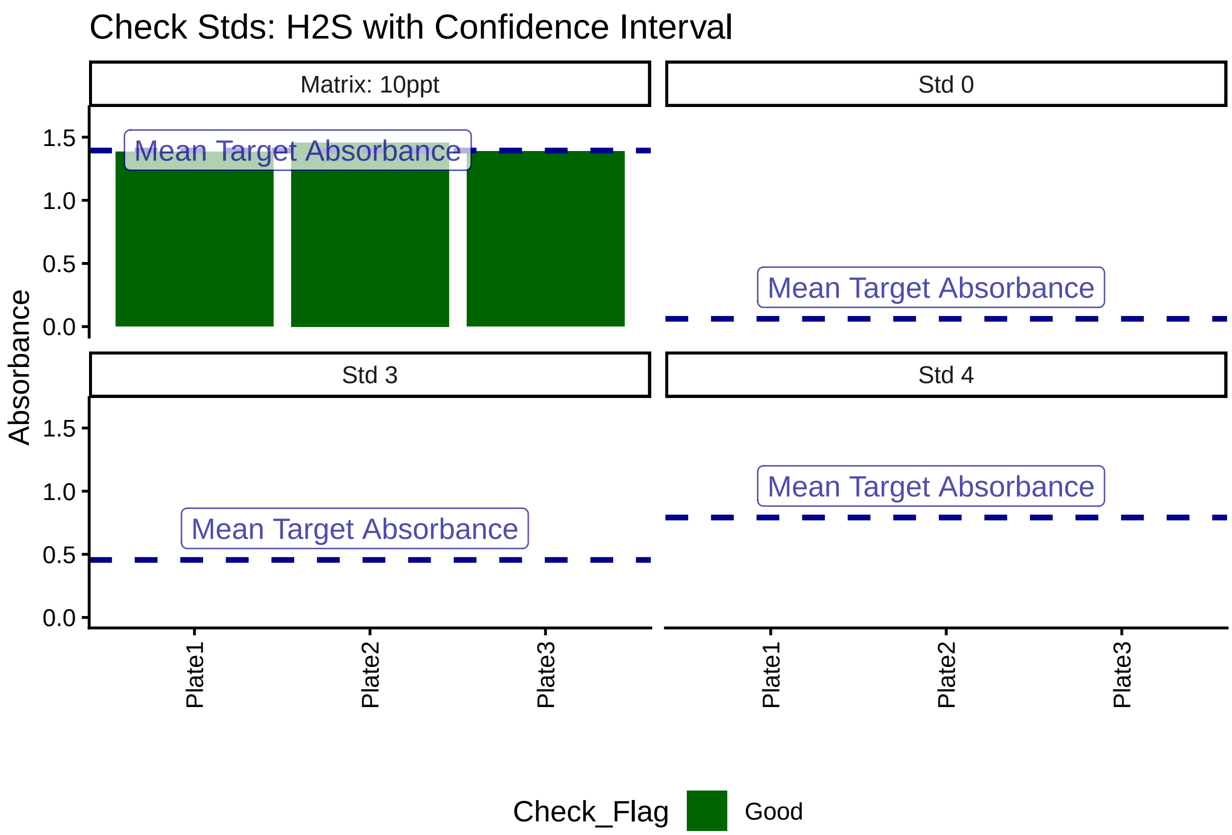
Plate	Curve_RMVD
Plate1	REMOVED
Plate2	REMOVED
Plate3	KEPT

Method Detection Limit



Compare Check Standards to Standards

```
## Warning in min(x): no non-missing arguments to min; returning Inf

## Warning in max(x): no non-missing arguments to max; returning -Inf

## Warning in min(x): no non-missing arguments to min; returning Inf

## Warning in max(x): no non-missing arguments to max; returning -Inf

## Warning in min(x): no non-missing arguments to min; returning Inf

## Warning in max(x): no non-missing arguments to max; returning -Inf

## Warning in FUN(X[[i]], ...): no non-missing arguments to max; returning -Inf
## Warning in FUN(X[[i]], ...): no non-missing arguments to max; returning -Inf
## Warning in FUN(X[[i]], ...): no non-missing arguments to max; returning -Inf
```



Display Any Check Flags

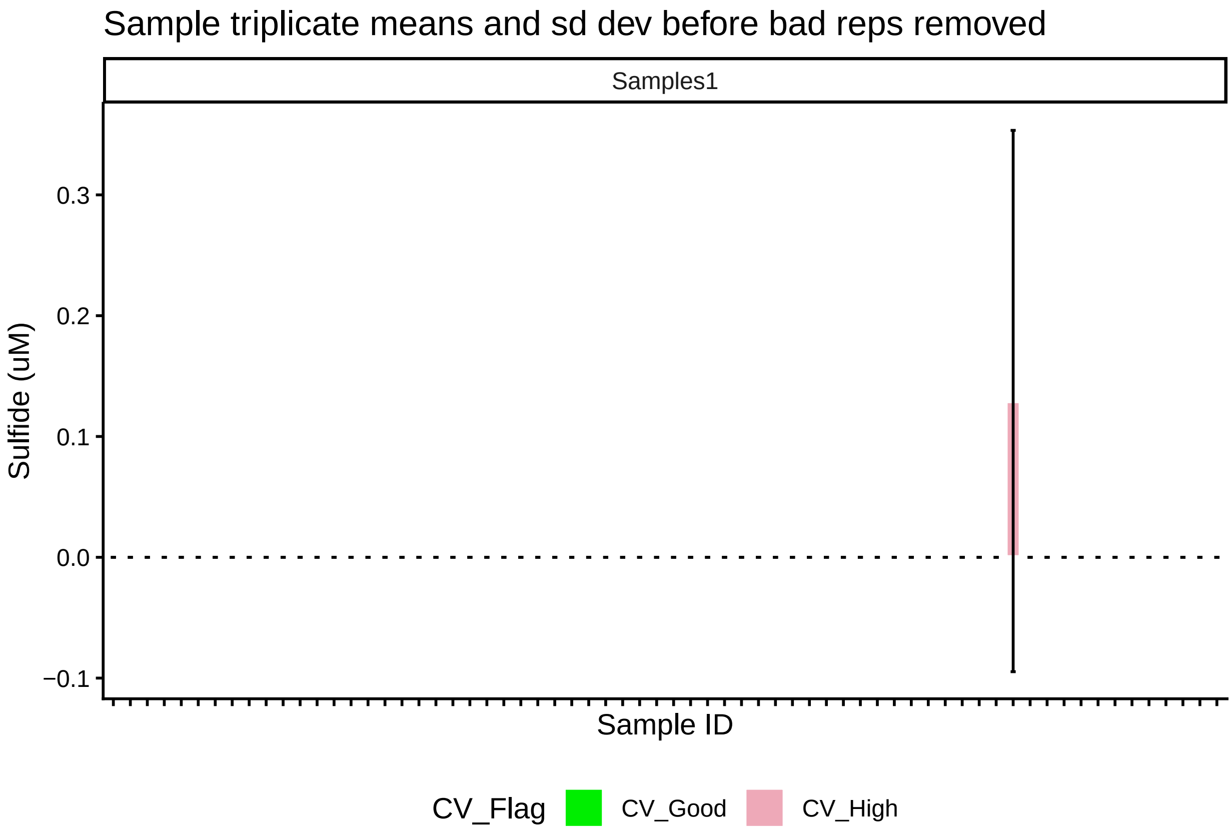
Plate	CHKs_Flag

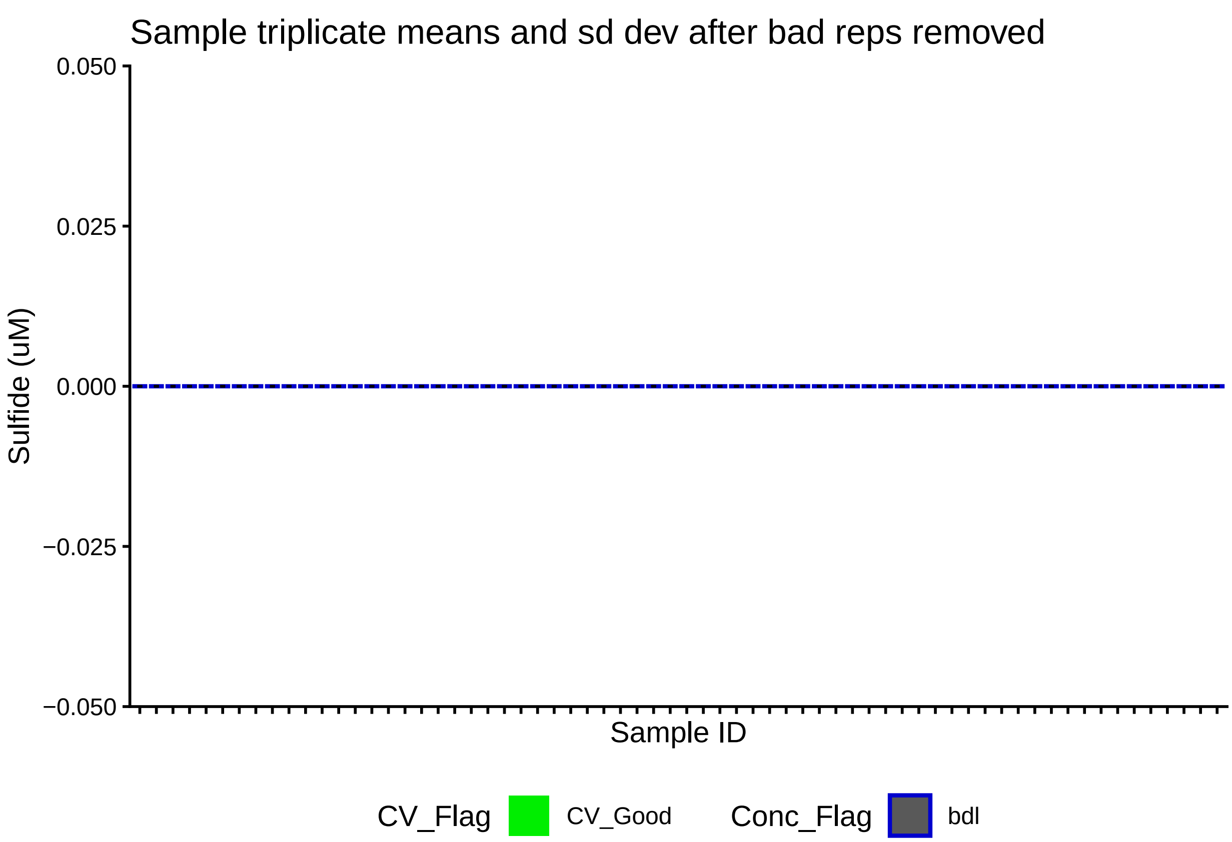
Plate	CHK_CV_Flags

##Calculate Sulfide Concentrations & Add Concentration Flags

##Remove High CV Samples and Average

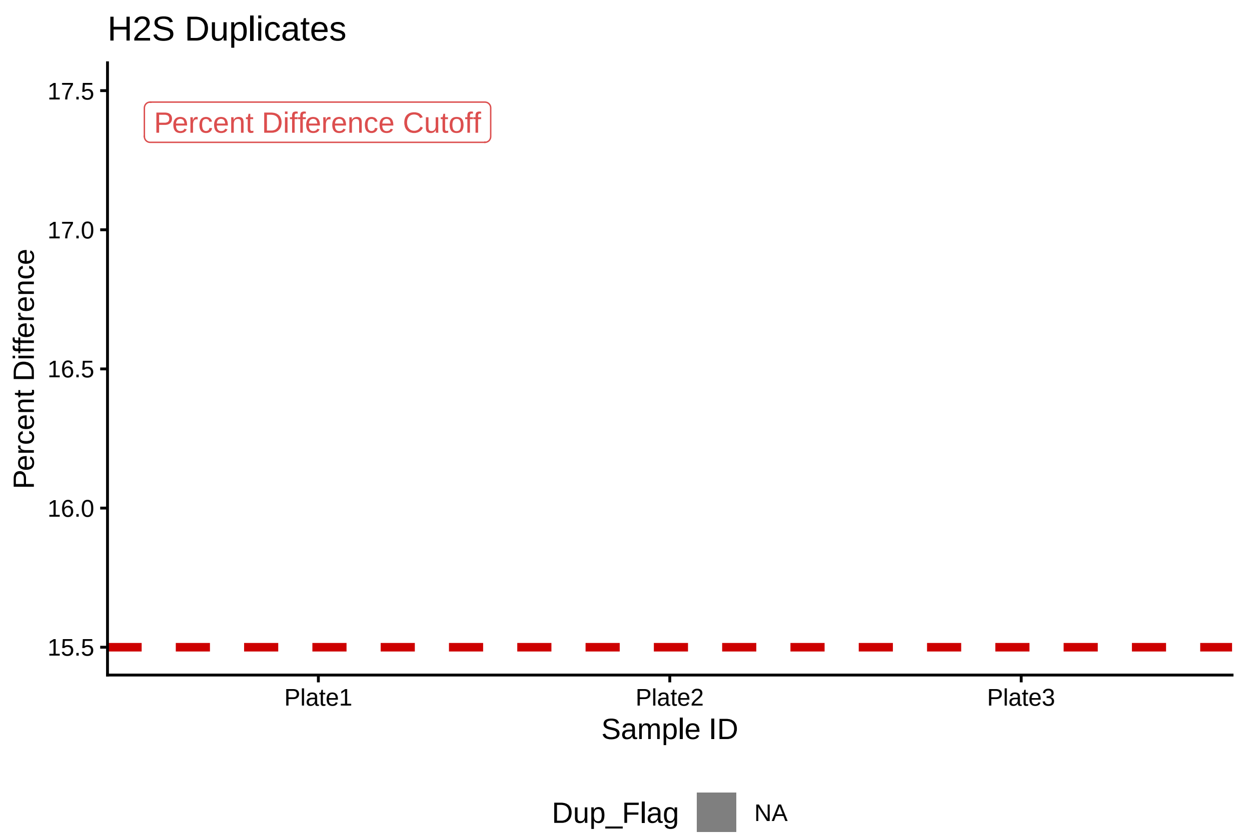
Plot Samples





Duplicate QAQC

```
## Warning: Removed 3 rows containing missing values or values outside the scale range
## (`geom_bar()`).
```

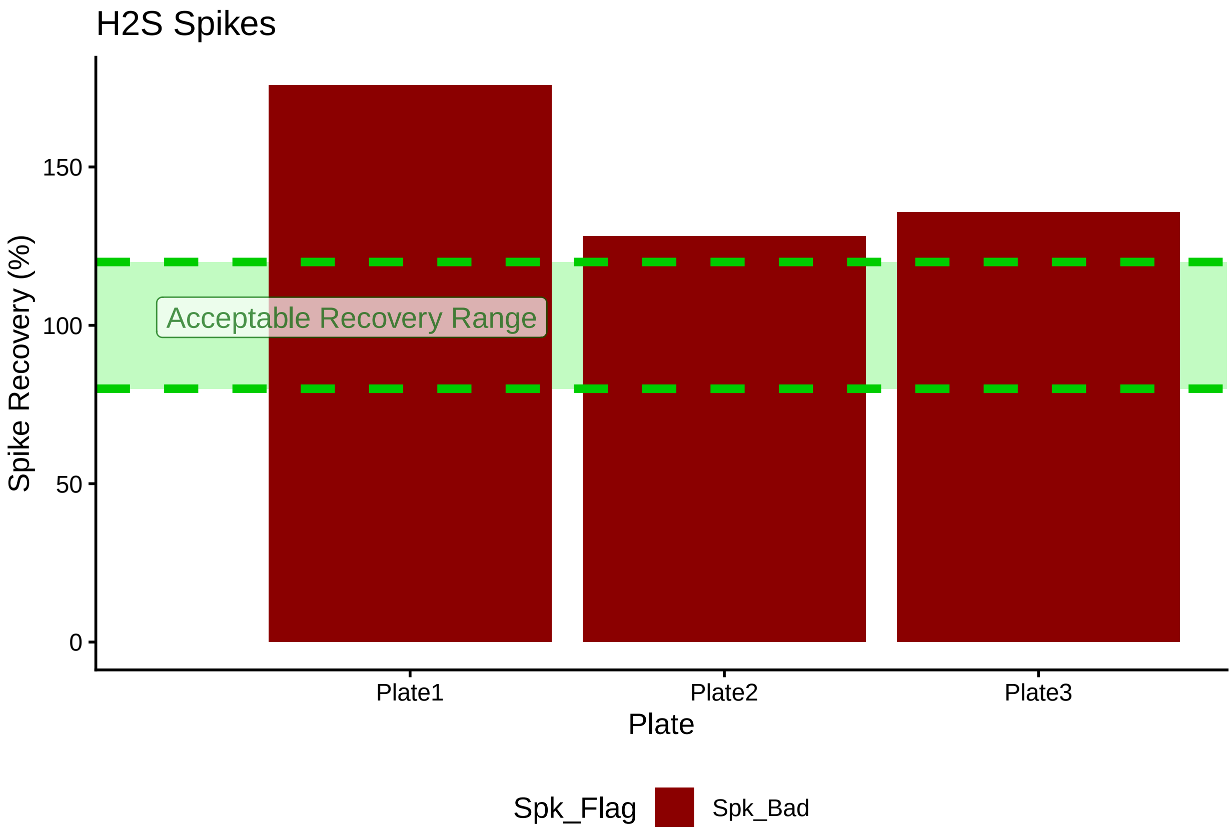


Display Dup Flags

```
## [1] ">60% Dups Pass"
```

Plate	Dup_Flag
Plate1	Dup_Good
Plate2	Dup_Good
Plate3	Dup_Good

Spike QAQC



Display Spike Flags

```
## [1] "Spikes Paired"
```

Plate	Spike_Flag
Plate1	Spk_Bad
Plate2	Spk_Bad
Plate3	Spk_Bad

```
## [1] "<60% of Spikes Pass, Rerun"
```

Check for reruns and replace

##Add Flags

##Remove Synoptic samples and rename TEMPEST samples

```
df_all_clean <- df_all_clean1 %>%
  filter(!str_detect(Sample_ID, "SWH|GCW|GWI|MSM"))

##TMP_C_C6_50_20260618 --> TMP_FW_PW_C6_50cm_S04_20260618
Sample_data_IDs <- df_all_clean %>%
  mutate(Plot = stringr::str_extract(Sample_ID, 'SW|FW|_C_|ESTUARY'),
         Plot = gsub("_", "", Plot),
         Grid = stringr::str_extract(Sample_ID, "B4|C3|C6|D5|E3|F4|F6|H3|H6|I5") ,
```

```
         Sample_Type = stringr::str_extract(Sample_ID, "SOURCE|RAINWATER"),
         Sample_Type = replace_na(Sample_Type, "PW"),
         Date = stringr::str_extract(Sample_ID, "[0-9]{8}"),
         Depth = stringr::str_extract(Sample_ID, "_15_|_30_|_50_"),
         Depth = gsub("_", "", Depth),
         Depth = paste0(Depth, "cm")) %>%
  mutate(Sample_ID = paste("TMP", Plot, "PW", Grid, Depth, "SO4", Date, sep = "_"))
```

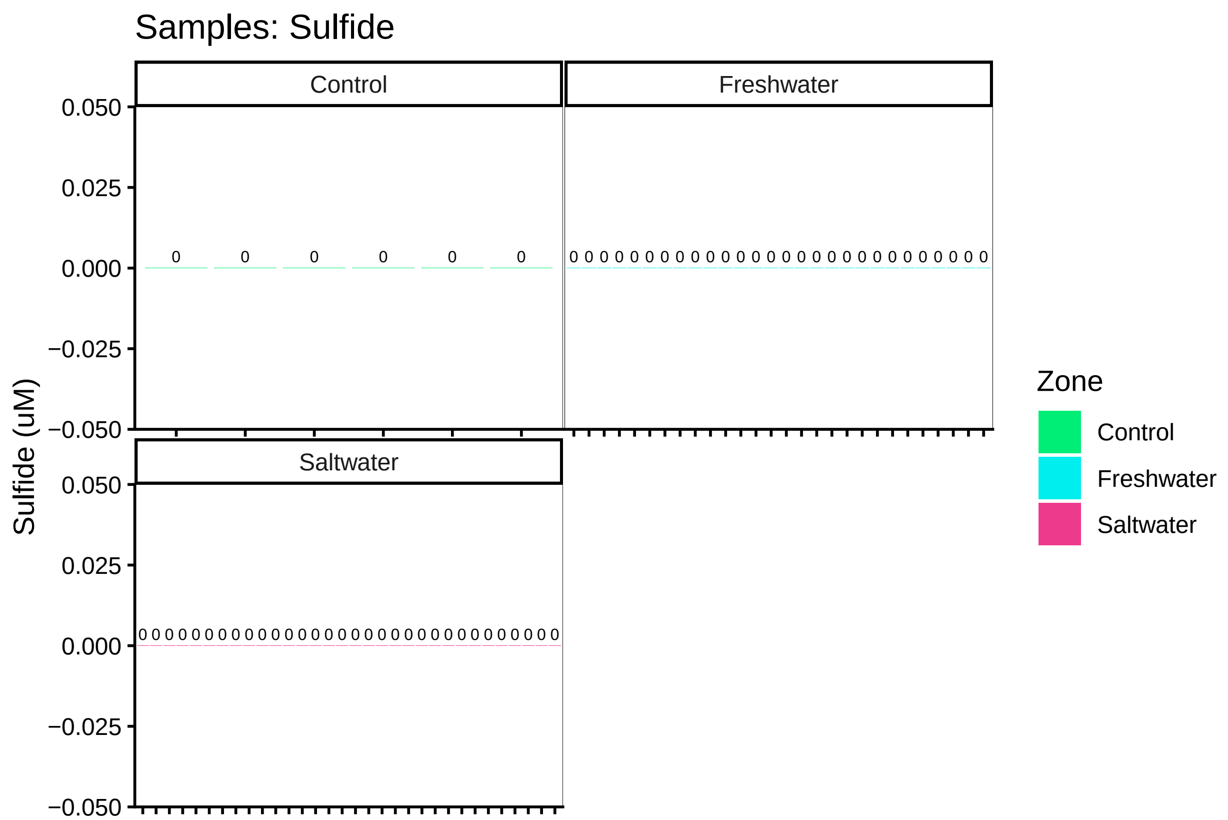
##Merge with Sample ID list to get actual sample IDs

##Pull in active porewater inventory

##Check samples against metadata

```
## ***All sample IDs are present in metadata.***
```

Visualize Data by Plot



##Export Data

END